Driving patient outcomes with the implementation of a social determinants of health (SDOH) network in an imbedded community pharmacy setting

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Background

Social determinants of health (SDOH) are the environmental conditions where people are born, live, work, and play, which affect their overall health, functioning, and quality of life outcomes.[1] Clark County, Nevada has a weighted Social Vulnerability Index (SVI) of 0.8635, which refers to the demographic and socioeconomic factors (such as poverty, lack of access to transportation, and crowded housing) that adversely affect communities.[2] This score indicates a high level of vulnerability, suggesting many patients in the Las Vegas/Clark County areas may have outstanding SDOH needs.[3]

Genoa Healthcare collaborated with a social care software company to expand SDOH services in Clark County, Nevada. This technology connects individuals in need with community-based organizations for SDOH support. The purpose of this study is to evaluate patient outcomes from the implementation of an SDOH network in an imbedded community pharmacy setting.

Objectives

Primary objective: Determine the impact of an SDOH network on participant medication persistence and adherence compared to non-participants (control).

Secondary objective: Quantify and categorize the SDOH cases resolved in the participant group.

Methods

Study design: Two-center, retrospective observational cohort study at Genoa Healthcare, deemed exempt by the Institutional Review Board of UnitedHealth Group. The participant group consisted of eligible patients who opted into the program with SDOH screening completed and case resolution, while the control group was comprised of eligible patients who were either unreachable or opted out of the program.

Inclusion criteria: New-to-pharmacy patients receiving HIV or behavioral health (BH) therapy at two Genoa pharmacies between March 2024 and December 2024 (required 6 months of dispensing data post-enrollment, ending June 31, 2025).

Data source: Data was collected from Genoa's prescription database and partner social care platform.

Statistical analysis: Persistence was evaluated between the groups by days on therapy (DOT) and persistence curve, while adherence was assessed by proportion of days covered (PDC) using propensity score matching (PSM) and Student's t-test. Alpha was set to 0.05 with a p-value of < 0.05 considered statistically significant. SDOH case resolution was performed through descriptive statistics.

Results

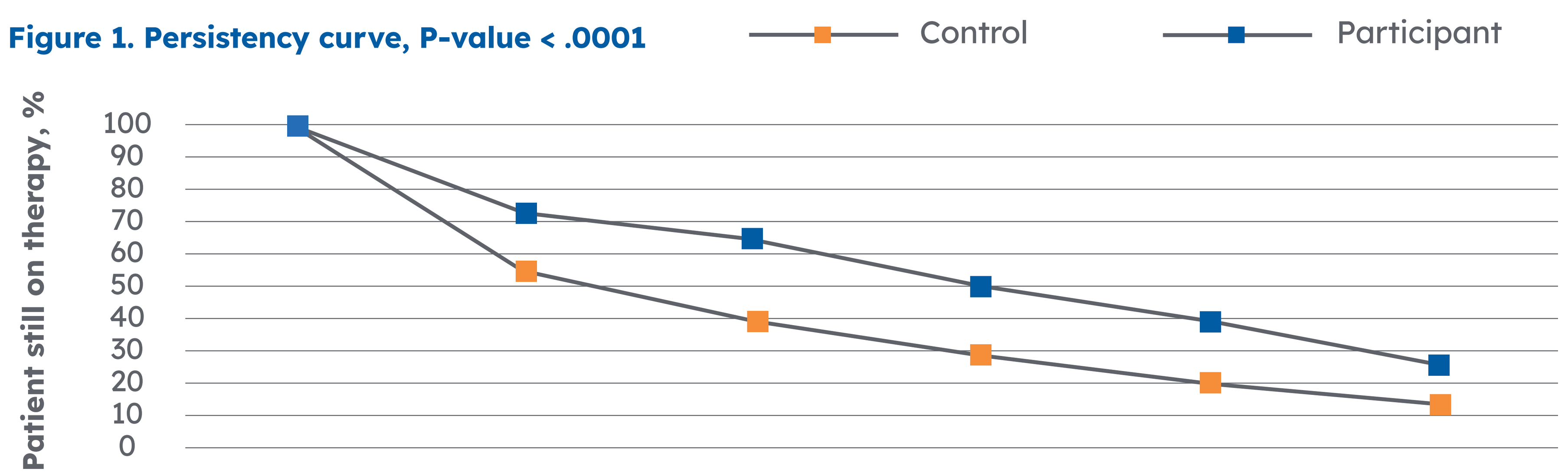
Table 1. Baseline patient characteristics, post-PSM

Variables	Control (N=112)	Participant (N=112)	P-value
Age, years (mean +/- SD)	36.3 ± 11.1	35.9 ± 11.9	0.79
First fill date (mean +/- SD)	8/3/2024 +/- 87	8/7/2024 +/- 73	0.73
Male, %	75.0	71.4	0.55
BH, %	58.0	59.8	0.78

Table 2. Patient outcomes

Outcomes	Control (N=112)	Participant (N=112)	P-value
PDC, % (mean +/- SD)	80 ± 20	81 ± 20	0.61
DOT, days (mean +/- SD)	147 ± 52	160 ± 47	0.06

Figure 1. Persistency curve, P-value < .0001



Fill 1	Fill 2	Fill 3	Fill 4	Fill 5	Fill 6
100.0	54.3	39.5	29.4	20.8	13.0
100.0	73.5	65.3	49.0	38.1	25.2

Figure 2. Participant SDOH referral status
March 2024–December 2024

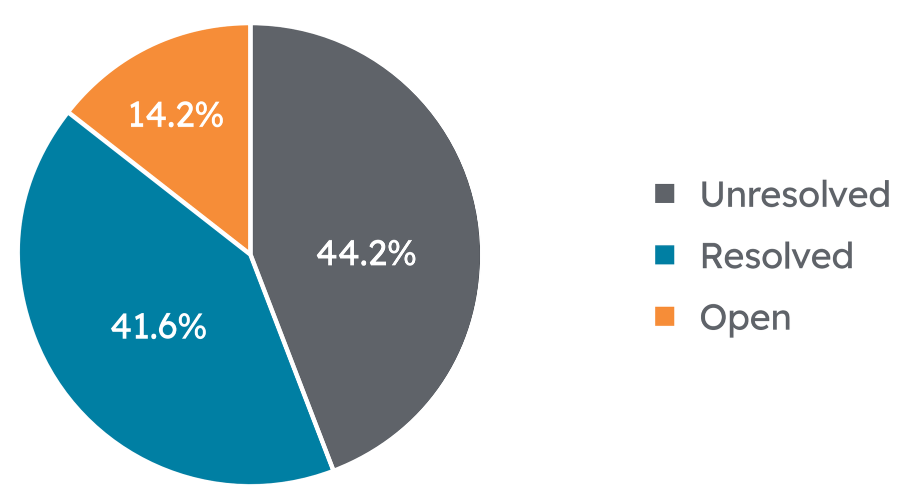
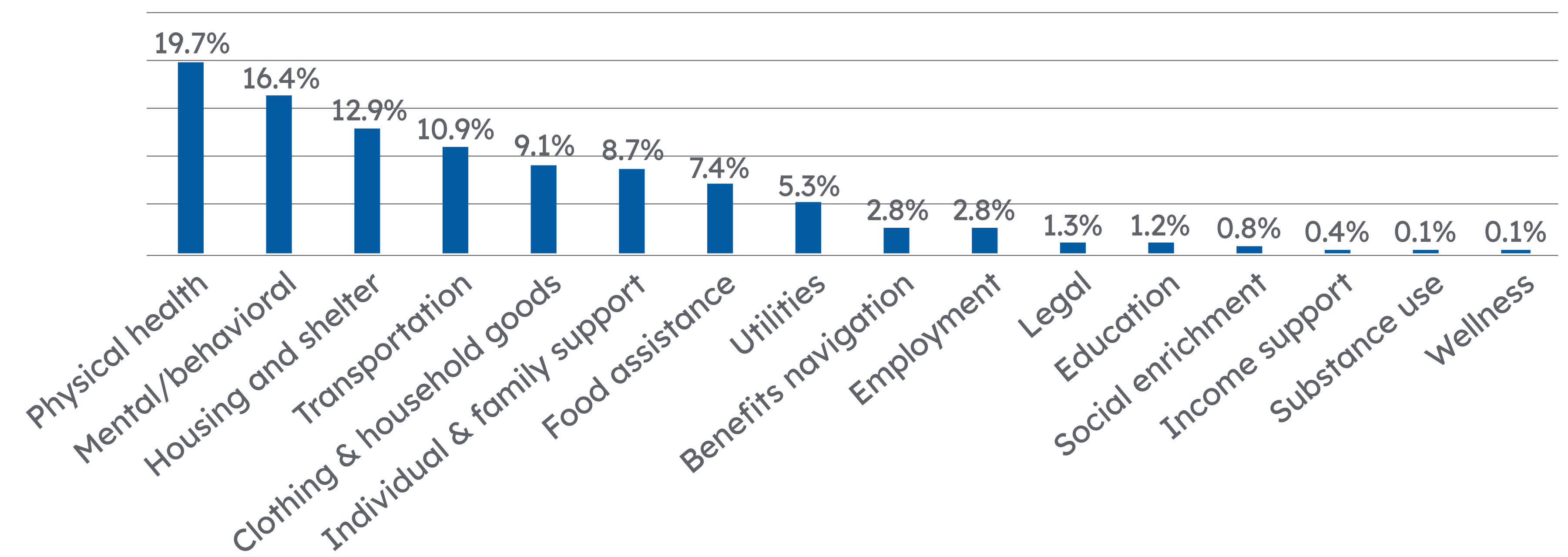


Figure 3. Resolved case volume by service type, average 4.4 referrals per participant



Discussion

- A partnership between a community pharmacy and a social care platform demonstrates that providing targeted SDoH support in the patient's local area can positively impact patient medication related outcomes.
- Clinically meaningful improved days on therapy were found in participant cohort, which can translate to an additional fill per year.
- Significantly more participants were on therapy at each fill compared to control.

Limitations

- This study is limited to patients at Genoa Healthcare in Las Vegas/Clark County and may not be generalizable to other healthcare settings.
- Patients in control cohort may have found local support outside of Genoa's program.

Future considerations

- Additional patient outcomes such as HIV viral response, BH-related hospital re-admission rates or other healthcare resource utilization could be considered to measure program impact.
- With positive outcomes found, this pilot could be scaled to larger populations.

References

1. Healthy People 2030, U.S. Department of Health and Human Services, Office of Disease Prevention and Health Promotion. Retrieved [date graphic was accessed], from https://odphp.health.gov/healthypeople/objectives-and-data/social-determinants-health
2. Centers for Disease Control and Prevention (CDC) and Agency for Toxic Substances and Disease Registry (ATSDR). (2024). SVI Interactive Map. Retrieved from https://www.atsdr.cdc.gov/place-health/php/svi/svi-interactive-map.html.
3. Centers for Disease Control and Prevention (CDC) and Agency for Toxic Substances and Disease Registry (ATSDR). (2024). Social Vulnerability Index. Retrieved from https://www.atsdr.cdc.gov/place-health/php/svi/index.html.

Disclosures/contact

Authors of this presentation have the following to disclose: Employees; stock holders of UnitedHealth Group

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